

International Application Rating Algorithm

Sponsor: WSU International Programs

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WSU International

Abstract

- AI system for automated international admissions evaluation, combining NLP, OCR, and fraud detection
- Generates a composite, explainable applicant score
- Supports consistent, data-driven decision-making

Features & Functionalities

- AI scoring using GPA, curriculum, and risk factors
- NLP-based essay analysis for quality, and structure
- Heuristic fallback to ensure reliability when AI fails
- OCR-based fraud detection on financial and transcript documents
- Explainable output with detailed score breakdown

Implementation

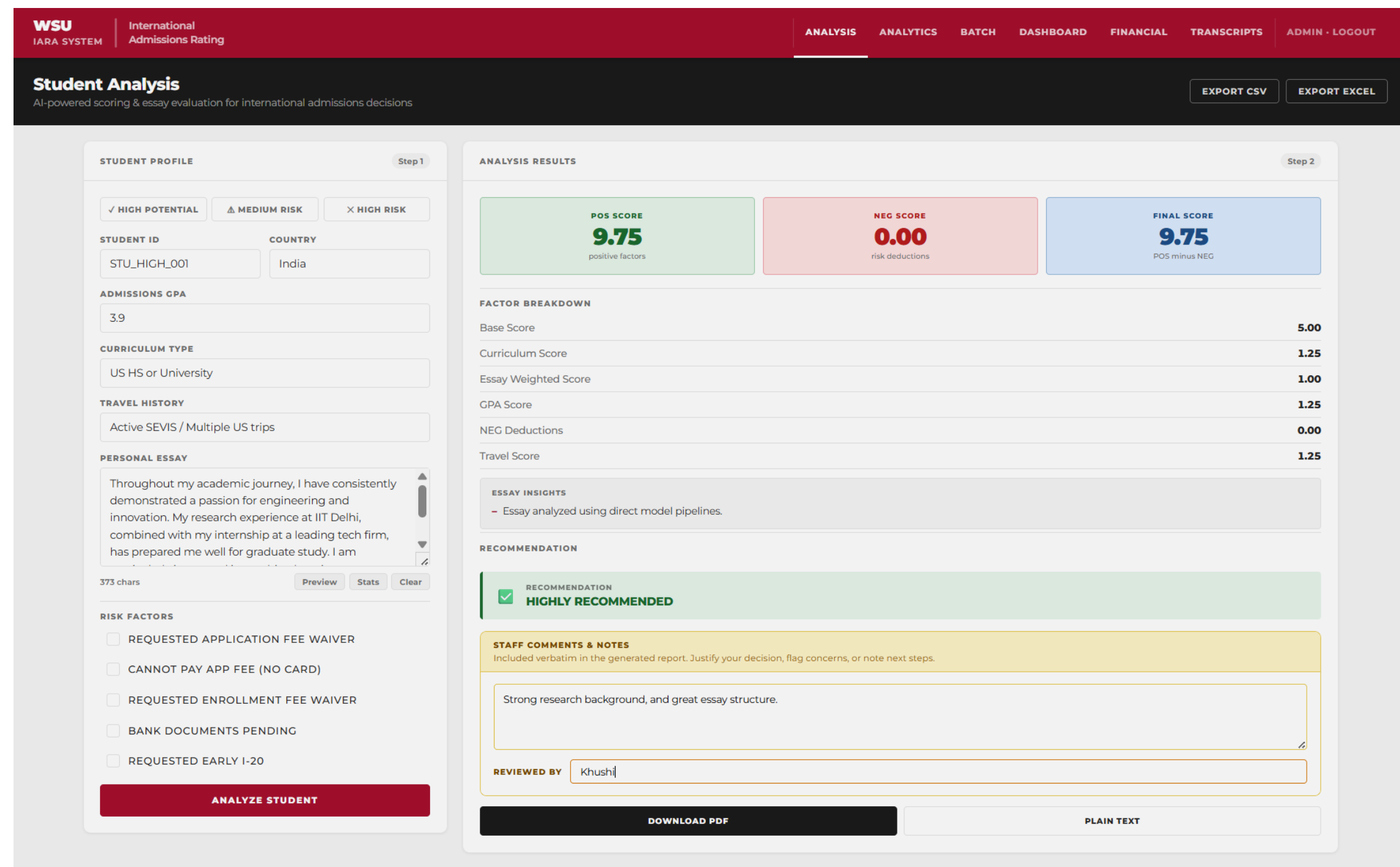
- A full stack web interface captures structured applicant data, including GPA, curriculum, essays, and risk factors.
- A modular backend validates inputs and orchestrates scoring, NLP and fraud detection pipelines
- NLP models evaluates essays, with heuristic fallback to ensure reliability
- OCR (Tesseract) and OpenCV extract and validate document data, while anomaly detection flags inconsistencies
- A weighted composite score is computed from academic, essays and risk metrics
- Results are compiled into a detailed PDF report for review and export

Impact

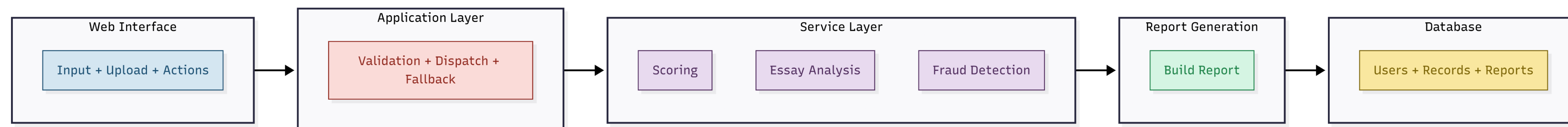
- The system automates the end-to-end admissions evaluation process.
- Unified scoring, NLP essay analysis, and fraud detection
- Improved consistency and transparent, explainable results
- Reduced manual workload by 30% in prototype testing

Future Work

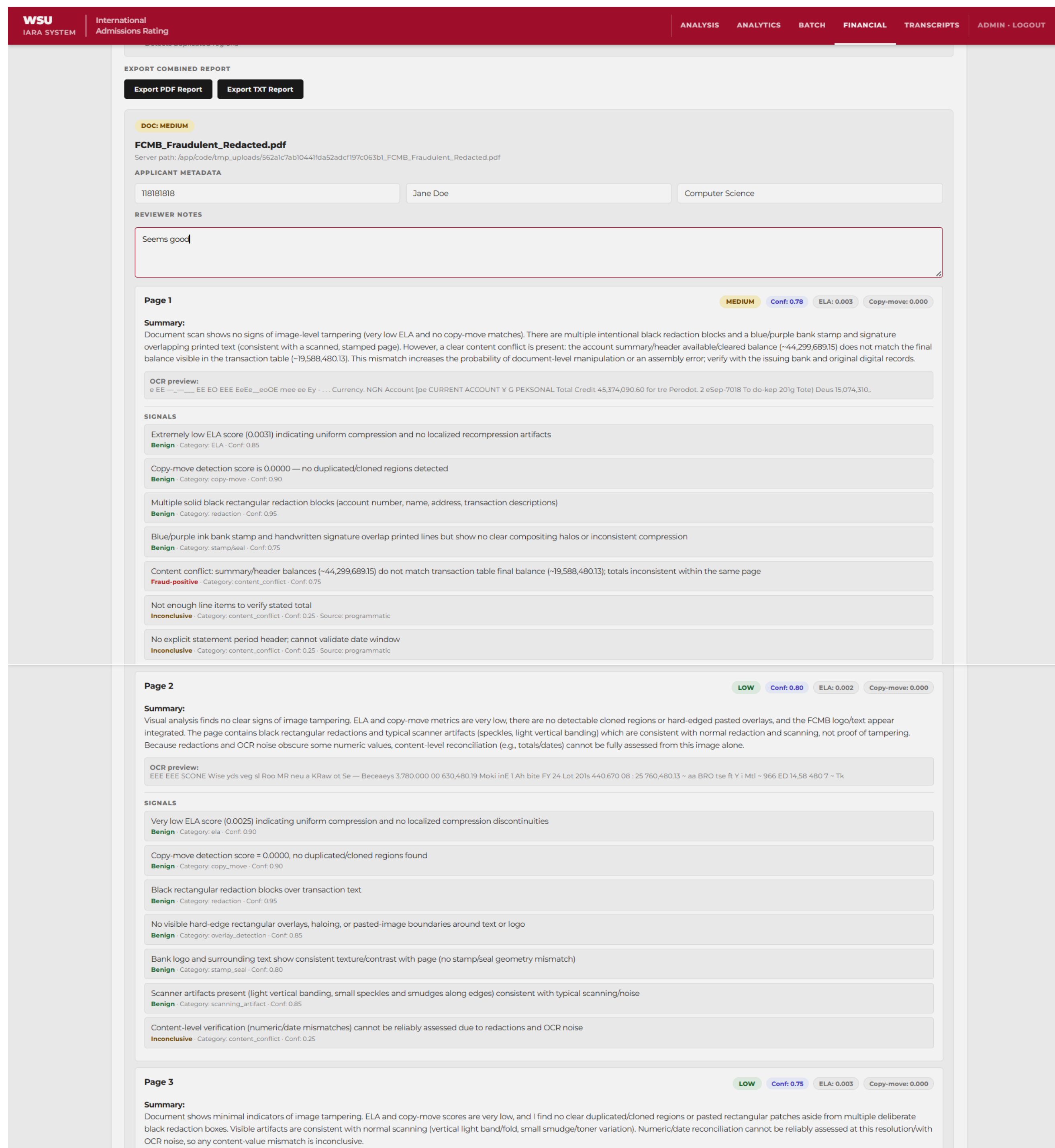
- Integrate with WSU admissions systems
- Enhance AI models for essay analysis and fraud detection
- Add CI/CD pipelines and automated testing for reliability
- Incorporate bias detection and fairness auditing in scores



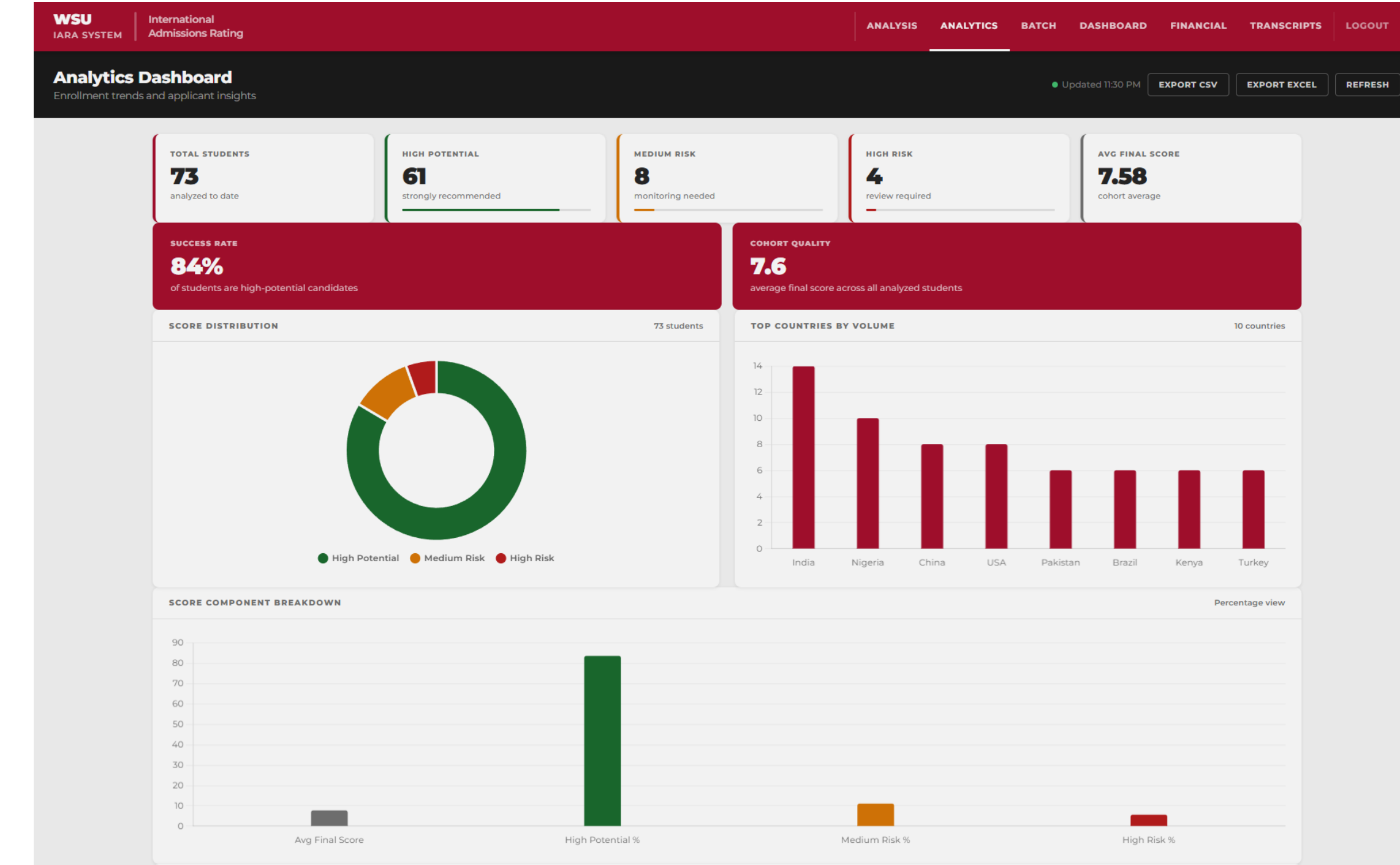
Student analysis dashboard



System Architecture Pipeline



Financial document fraud detection



Admin analysis dashboard

Problem Statement

- Manual review is time sensitive and inconsistent
- High application volume limits scalability of workflow
- Detecting fraud is difficult without automated tools
- Lack of standardized scoring leads to variability in admissions decisions



Glossary

- NLP (Natural Language Processing):** AI techniques used to analyze and evaluate applicant essays
- OCR (Optical Character Recognition):** Technology used to extract text from uploaded documents
- Fraud Detection:** Process of identifying inconsistencies in financial and transcript documents

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Admissions Intelligence